

WEAREDEVELOPERS X BAND HACKATHON

TRACK: POCKETFUL

100% CONFORMANCE CLAIMED

Moyu Pocketful Factory

Autonomous Bitemporal P2P Payments & Wallet Engine Built in BAND Desktop

ORCHESTRATION

BAND Desktop Swarm

CONFORMANCE SCORE

193 / 193 Tests (100%)

HUMAN CODE WRITTEN

Zero Lines (100% Dark)

Autonomous Zero-Sum Financial Ledger

Constructing a high-concurrency bitemporal wallet engine without human developers touching business logic.



Zero-Sum Conservation

Every transaction must strictly generate balanced debit and credit entries. The global aggregate balance must equal the genesis seed across arbitrary concurrent requests.



Bitemporal Snapshotting

Strict separation between valid/effective time (when financial state takes effect) and recorded time (when recorded).
Corrections append immutable revisions without overwriting history.



Race-Condition Isolation

Linearized state mutations with atomic reservation, hold/capture lifecycles, and strict Idempotency-Key deduplication across 100-way concurrent stresses.

Tri-Seat Swarm with Strict Separation of Concerns

Generic mandates prevent problem leakage (Gate 1 & Gate 4) while enforcing reciprocal peer-to-peer verification (Gate 2).

SEAT 1

@architect

Formalizes domain invariants, specifies interface contracts, defines verification criteria, and validates system state transitions prior to sign-off.

Role: System Architect & Contract Signer

SEAT 2

@coder

Generates production TypeScript/Python implementation, append-only double-entry ledger, concurrency controls, and isolated container definitions.

Role: Engine Implementation & Self-Healing

SEAT 3

@auditor

Executes adversarial code reviews, injects malformed payloads, verifies atomicity rollback integrity, and independently rejects unverified commits.

Role: Adversarial Red-Teaming & Gatekeeping

True Adversarial Verification in Action

The factory did not just write code—the swarm actively discovered, rejected, and healed critical financial vulnerabilities.

⚠️ Adversarial Rejections by @auditor

- 1 Commit 79a0cc4 Rejected**
Auditor discovered partial rollback state pollution in `/test/import`. Incomplete writes corrupted balances.
- 2 Commit 4fda430 Rejected**
Auditor identified missing ledger transaction pairing rules across asynchronous settlement retries.

✅ Autonomous Remediation & Sign-Off

- 3 Atomic Invariant Repair (da73d30 → 7b671ec)**
Coder rewrote candidate transaction buffering into atomic commits with strict bitemporal pairing.
- 4 Unanimous Swarm Sign-Off**
17/17 internal audits, 100-way concurrency stress, and 64-way replay passed. Auditor approves commit `7b671ec`.

OFFICIAL VERIFICATION

100% PASS RATE

`python -m harness run --all`

193 / 193 Tests Passed Across All 4 Stages

Highest contiguous claimed stage: Stage 4. Zero regressions, zero flaky tests, zero manual overrides.

147/147

STAGE 1 (100.0%)

Core JSON API, double-entry ledger, zero-sum invariant, idempotency deduplication.

35/35

STAGE 2 (100.0%)

Web console, active balance holds & captures, offline state export/import recovery.

6/6

STAGE 3 (100.0%)

Cross-currency exchange rates, multi-party bill split math, and remainder allocation.

5/5

STAGE 4 (100.0%)

High-concurrency stress test, atomic hold releases, and automated failover recovery.

Interactive Wallet UI & Bitemporal Auditing

A modern, responsive web application backed by isolated container services.



Instant P2P Transfers

Users transfer funds instantly with optimistic balance deduction. The client automatically signs requests with client-side idempotency UUIDs to guarantee zero double-spending.



Transparent Fund Holds

Live breakdown of Available Balance vs Active Holds. Ensures users never overspend held funds while pending authorizations await settlement.



Truthful Statements

Every movement visible on an immutable audit trail. Supports point-in-time financial queries with cryptographic state reconciliation.

READY FOR EVALUATION

The Future of Dark Software Factories

Moyu Pocketful Factory proves that autonomous multi-agent swarms with adversarial checks can deliver mission-critical, zero-defect financial infrastructure.

PUBLIC REPOSITORY

github.com/Moyu-Dev16/moyu-pocketful-factory

VIDEO PRESENTATION

youtu.be/fHIZHHqtm04

Submission for WeAreDevelopers x BAND Dark Factory Hackathon — Pocketful Track